

GOODNESS-OF-FIT TEST FOR CONDITIONAL MOMENT RESTRICTION MODELS

A THESIS

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Abstract

This thesis provides a test that performs systematic model validation under the moment restriction structure by a series of equivalent tests for the goodness-of-fit of regression models with respect to conditional-moment restrictions. Furthermore, this thesis tries to eliminate the first order estimation effect on the test statistics by changing the Gaussian kernel that is used in the testing procedure. Using a kernel adjustment method to decouple the test statistics from the estimated model parameters, which improves the validity and robustness of the conditional moment restrictions hypothesis testing.

KEY WORDS: Conditional Moment Restriction Models, goodness-of-fit test, estimation effect

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Chapter 1

Introduction

1.1 Overview of Conditional Moment Restriction Models

Conditional moment restriction models are fundamental to the formulation of a wide range of econometric models. It also plays a central role in identifying and estimating the parameters of interest. These models are constructed according to the specific conditional statistical moments (i.e. conditional expectations, variances, or higher-order moments) as functions of observed covariates.

Moreover, they are not dependent on assumptions about the underlying probability distribution of the observed data which provides a flexible and semiparametric modeling framework. Due to the generality of this approach, conditional moment restriction models can be applied to a large array of empirical contexts.

The models based on this framework also have the advantage that, due to their low dependence on parametric assumptions, these models do not impose strong constraints on the distributional assumptions of error terms and on functional relationships among variables, thus being especially fitted for the description of complex economic dynamics and for accommodating heterogeneous structures of data. This flexibility is particularly useful in situations where the data-generating process is latent or complex.

Thus, restrictions on conditional models provide a natural basis for model specification tests. These restrictions represent the dependence across the variables in terms of their conditional moments (conditional means, variances, covariances and higher-order moments like skewness and kurtosis). Hypothesis tests based on these moments are therefore used to test whether the conditions implied by the model hold when compared to the empirical distribution of the observed data.

Thus, the validity of conditional moment restrictions —is a basic assumption of the correctness of model specifications. Model misspecification, that leads to the breakdown of these moment conditions, then causes an inferential chain reaction that eventually jeopardizes the integrity of

empirical analysis:

Inconsistency of Parameter Estimators: Violation of moment restrictions may causes parameter estimators inconsistent, manifesting the systematic deviations of estimated coefficients from their unknown population true value.

Invalidity of Standard Error Estimation: Under the violation moment restrictions, traditional estimators of the asymptotic variance-covariance matrix lose statistical accuracy which undermines the reliability of reported standard errors.

Distortion of Asymptotic Distributions: Estimators may have asymptotic bias, while the central limit theorem underpinning the asymptotic normality of estimators, necessarily collapses under misspecified moment restrictions.

1.2 Goodness-of-Fit Testing under the ICM Framework

This thesis adopts the Integrated Conditional Moment (ICM) framework (**Bierens,1982**) [1] to highlight how a method based on solid theoretical grounding can be transformed into a goodness-of-fit testing methodology. ICM revolves around conditional moment restrictions, the idea being to form test statistics that check whether a model is appropriately specified. Its approach is based on a number of rigorous analytical building blocks.

Generalized Residual Function

In the Goodness-of-Fit Testing framework, this thesis initiates the ICM methodology with a generalized residual function that it derives from conditional expectations. It also underlies parameter estimation and hypothesis testing. It measures the gap between the actual results and their conditional means and hence connects the observed variables and the latent relationship via instrumental variables.

Transformation from Conditional to Unconditional Moments

The type of weighting functions used in this transformation remove the dependence on the conditioning variables. Not only is this procedure conducive to the construction of the moment conditions, but it is also critical to guarantee that the resulting test statistics are invariant to possible nuisance parameters.

Integral-Transform Test and Empirical Process Approximation

Using the Integral-Transform Test by **Miguel A., Delgado, Manuel A. Domínguez and Pascal Lavergne in 2006** [3] to derive low-dimensional projections of the transformed moment conditions onto scalar-valued, test statistics that solely depend on the observed data and model specification. At the same time, an empirical process—justified by the Central Limit Theorem—is built to approximate the sampling distribution of the test statistic computed under the

null hypothesis. This is a key step to obtain the asymptotic distribution needed for statistical inference (i.e., critical values, p-values, etc.).

Chapter 2

Literature Review

The **Generalized Method of Moments (GMM)**, suggested by **Hansen** (1982)[4], based on a finite sample of unconditional moment conditions $\mathbb{E}[g(X, \theta)] = 0$. A central issue in this case, as in many over-identified problem (i.e. one in which the number of moment conditions is in excess of the number of parameters), is that testing must be conducted to assess the joint validity of the moment restrictions. **Hansen's J-test**[4] is the usual way to do this for validation, to test that the chosen moment conditions do in fact hold at the point estimate. But there are two serious sins in that approach. 1)Finite-Dimensional Constraint: The test is intrinsically finite-dimension as it tests finitely many moment conditions and cannot test an infinite number of conditional moment conditions in infinite-dimension. 2)Vulnerability to Model Misspecification: The established test has an asymptotic size equal to that under the null, if the misspecification is far from the span of the given moment conditions, with this possibility resulting in incorrect inference.

To address this issues, **Newey** (1983)[8] and **Tauchen** (1985)[11] suggest constructing a methodological framework that selects a finite set of weighting functions to construct a subset of **unconditional moment restrictions (UMR)** with a finite number of weighting functions and then performing over-identification testing in the GMM framework. However, there are also few consequences of this procedure upon the asymptotic correctness and the empirical robustness.

Critically, these limitations call for a nonparametric conditional moment test for a full model validation outside the finite dimensional framework.

The groundbreaking **Integrated Conditional Moment (ICM)** test proposed by **Bierens** (1982) [1] has transformed conditional moment analysis by constructing a continuum of unconditional moment restrictions (UMR) through a parameterized family of weighting functions.

A key difficulty in the hypothesis testing is that the original test statistic is a conditional expectation (in ICM tests). Conditional expectations almost always have to do with dealing with a random variable given some conditions, which is complicated and the testing procedure with nuisance or unknown parameters. So directly employing an some conditional expectation-based

statistic for testing is challenging in these cases because it may depend on unobserved parameters or certain model complexities.

To adjust for this, there exists a known technique that defines an appropriate real-valued function $w(\cdot)$, allowing the transition of the conditional expectation to an unconditional one. That is, once an appropriate function $w(\cdot)$ is chosen, the statistic's form can change from a conditional expectation that depends on specific parameters to one that is unconditional. This transformation makes the hypothesis testing process simpler and more relevant.

It is quite complicated to prove theoretically how this transformation is equivalent, but **(STINCHCOMBE, WHITE 1998)** [10] suggest that weighting functions can generate consistent tests by carefully choosing the function restrict to some particular class of functions. Most notably,

$$w_P(X, x) = G(X'x)$$

with $G(\cdot)$ the logistic cumulative distribution function or the exponential employed by **(Bierens 1982)** [1] making it a good tool for the transformation.

This framework of **Bierens's** ICM generates an infinite-dimensional system of moment conditions and formulates a quadratic-form test statistic in integral space, exhaustively spanning all potential directions of model misspecification. The test achieves asymptotic omnibus power, characterized by its rejection probability converging to 1 as the sample size $n \rightarrow \infty$ under global alternatives. Under the null hypothesis, the test statistic converges weakly to an infinite-dimensional chi-square process, necessitating critical value approximation through simulation techniques (e.g., wild bootstrap) or Monte Carlo methods. This resolves the intrinsic incompatibility between infinite-dimensional moment constraints and finite computational implementations, while providing a robust diagnostic tool for high-dimensional and nonparametric models, including those integrating machine learning where traditional moment-based tests are inapplicable. However, computational challenges arise in empirical practice. When the baseline data X is high-dimensional, the integration domain experiences the curse of dimensionality, rendering numerical quadrature or Monte Carlo sampling computationally prohibitive.

Li and Song (2022) [6] propose characterizing test statistics through the **Maximum Moment Restriction (MMR)** framework. This methodology employs a kernel function to construct the statistic directly via kernel evaluations, circumventing explicit integration. The closed-form representation requires only the computation of a kernel matrix derived from sample pairs. When utilizing a universal kernel (e.g., Gaussian, polynomial), the MMR statistic uniquely characterizes the original conditional moment constraint $E[\varepsilon | X] = 0$. By leveraging the reproducing property and denseness property of universal kernels in **reproducing kernel Hilbert spaces (RKHS)**, the MMR statistic vanishes if and only if all conditional moment restrictions are satisfied, systematically avoiding the curse of dimensionality inherent to classical nonparametric tests. The closed-form structure ensures computational efficiency while being scalable to high-dimensional data environments. This framework establishes a unified paradigm for conditional

moment testing that harmonizes statistical rigor with computational scalability.

Chapter 3

Methodology

3.1 von Mises statistic (V-Statistics)

For sections 3.1 and 3.2, all the details are thoroughly covered in Chapter 5 of the book by Serfling [9].

Let $X_1, X_2, \dots$ be independent observations on a distribution F . For some function $h = h(x_1, \dots, x_m)$, called a "kernel." Without loss of generality, it may be assumed that h is symmetric. For, if not, it may be replaced by the symmetric kernel.

$$\frac{1}{m!} \sum_p h(x_{i_1}, \dots, x_{i_m})$$

Consider a "parametric function" $\theta = \theta(F)$ for which there is an unbiased estimator. That is, $\theta(F)$ may be represented as

$$\theta(F) = E_F \{h(X_1, \dots, X_m)\} = \int \dots \int h(x_1, \dots, x_m) dF(x_1) \dots dF(x_m),$$

where $\sum_p$ denotes summation over the $m!$ permutations $(i_1, \dots, i_m)$ of $(1, \dots, m)$.

For any kernel h , the corresponding von Mises statistic (V-Statistics) for estimation of θ on the basis of a sample $X_1, \dots, X_n$ of size $n \geq m$ is obtained by averaging the kernel h symmetrically over the observations:

$$\begin{aligned} V_n &= \frac{1}{n^m} \sum_{i_1=1}^n \dots \sum_{i_m=1}^n h(X_{i_1}, \dots, X_{i_m}) \\ &= \theta(F_n) \end{aligned}$$

Intuitive understanding of V-Statistics is that it is about plugging the empirical cumulative distribution function in a statistical functional instead of a true cumulative distribution function to come up with a estimator and it is also a generalization of sample moments.

3.2 U-Statistics

Defination

Based on above, U-Statistics closely related V-Statistics. Intuitive thinking is that to make V-statistics unbiased, remove the duplicates in $i_1, \dots, i_m$ to make $h(X_{i_1}, \dots, X_{i_m})$ are independent by defining the form of U-Statistics in the following:

$$U_n = \frac{1}{\binom{n}{m}} \sum_c h(X_{i_1}, \dots, X_{i_m}).$$

$\sum_c$: means the sum of all combinations except the duplicates. Then explore further properties of U-Statistics

The Variance Of U-Statistic

With the following settings:

- (i) Define: $h_c(X_1, \dots, X_c) = \mathbb{E}_F[h(X_{i_1}, \dots, X_{i_m}) \mid X_1, \dots, X_c]$.
- (ii) Also : $\hat{h} = h - \theta$, $\hat{h}_c = h_c - \theta$.
- (iii) Denote that: $\sigma_c^2 = \text{Var}_F[h_c(X_1, \dots, X_c)]$.

Thus: $\mathbb{E}_F[h_c(X_1, \dots, X_c)] = \mathbb{E}_F[\mathbb{E}_F[h(X_{i_1}, \dots, X_{i_m}) \mid X_1, \dots, X_c]] = \mathbb{E}_F[h(X_{i_1}, \dots, X_{i_m})] = \theta$.
Then, $U_n - \theta = \binom{n}{m}^{-1} \sum_c \hat{h}(X_{i_1}, \dots, X_{i_m})$. To approach the variance of U-statistic:

$$\text{Var}_F(U_n) = \binom{n}{m}^{-1} \sum_{c=0}^m \binom{m}{c} \binom{n-m}{m-c} \sigma_c^2.$$

With the properties:

1. $\frac{m^2}{n} \sigma_1^2 \leq \text{Var}_F(U_n) \leq \frac{m}{n} \sigma_m^2$.
2. $(n+1) \cdot \text{Var}_F(U_{n+1}) \leq n \cdot \text{Var}_F(U_n)$.
3. $\text{Var}_F(U_n) = (\frac{m^2 \sigma_1^2}{n} + O(n^{-2}))$ as $n \rightarrow \infty$

The Projection of U-Statistic On The Basic Observations

$$\hat{U}_n = \sum_{i=1}^n \mathbb{E}_F\{U_n \mid X_i\} - (n-1)\theta.$$

$$\hat{U}_n - \theta = \sum_{i=1}^n \mathbb{E}_F[U_n - \theta \mid X_i]$$

$$\begin{aligned}
&= \sum_{i=1}^n \left[\frac{1}{\binom{n}{m}} \sum_c \mathbb{E}_F \left[h(X_{i_1}, \dots, X_{i_m}) - \theta | X_i \right] \right] \\
&= \sum_{i=1}^n \left[\frac{\binom{n-1}{m-1}}{\binom{n}{m}} (h_1(X_i) - \theta) \right] \\
&= \sum_{i=1}^n \left(\frac{m}{n} h(X_i) - \theta \right) \\
&= \frac{m}{n} \sum_{i=1}^n \hat{h}_1(X_i)
\end{aligned}$$

Based on the result above, it can be seen that:

$$U_n - \hat{U}_n = \frac{1}{\binom{n}{m}} \sum_c H(X_{i_1}, \dots, X_{i_m}),$$

where:

$$H(X_1, \dots, X_m) = h(X_1, \dots, X_m) - \hat{h}_1(X_1) - \dots - \hat{h}_m(X_m) - \theta.$$

Note that:

$$\mathbb{E}_F[H] = \mathbb{E}_F[H(X_1, \dots, X_m)] = 0.$$

With the properties:

(i) $\mathbb{E}_F[(U_n - \hat{U}_n)^2] = O(n^{-1})$, as $n \rightarrow \infty$.

(ii) Almost Sure Representation of a U-statistic as a mean of i.i.d.'s :

Let ν be an even integer. Suppose $\mathbb{E}_F[h^\nu] < \infty$. Define: $U_n = \hat{U}_n + R_n$. Then, for any $\delta > \frac{1}{\nu}$, with probability 1:

$$R_n = O(n^{-1}(\log n^\delta)), \quad n \rightarrow \infty.$$

Generalized result

For more than one given basic observation in the conditional term:

$$\begin{aligned}
\hat{U}_n - \theta &= \sum_{1 \leq i_1 < \dots < i_c \leq n} \mathbb{E}_F\{U_n | X_{i_1}, \dots, X_{i_c}\} - \binom{n}{c} \theta. \\
&= \frac{m(m-1) \dots (m-c+1)}{n(n-1) \dots (n-c+1)} \sum_{1 \leq i_1 < \dots < i_c \leq n} \hat{h}_c(X_{i_1}, \dots, X_{i_c}).
\end{aligned}$$

Additional Theorem on Asymptotic Distribution

If $\mathbb{E}_F h^2 < \infty$ and $\zeta_1 = 0 < \zeta_2$, then:

$$n(U_n - \theta) \xrightarrow{d} \frac{m(m-1)}{2} Y,$$

where Y is a random variable of the form:

$$Y = \sum_{j=1}^{\infty} \lambda_j (\chi_{1j}^2 - 1),$$

and $\chi_{1j}^2, \chi_{12}^2, \dots$ are independent χ^2 variates. That is, Y has the characteristic function:

$$\mathbb{E}_F(e^{itY}) = \prod_{j=1}^{\infty} (1 - 2it\lambda_j)^{-1/2} e^{-\lambda_j}.$$

3.3 Application of Bochner's Theorem

(Bochner) [2]. A continuous function $\varphi : \mathbb{R}^d \rightarrow \mathbb{C}$ is positive definite if and if is the Fourier transform of a finite nonnegative Borel measure Λ on $\mathbb{R}^d$:

$$\varphi(t) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-it^\top \omega} d\Lambda(\omega).$$

for $t \in \mathbb{R}^d$.

In this thesis, normalize φ such that $\varphi(0) = 1$, Λ is a normal distribution of the form $(2\pi/\sigma^2)^{-d/2} e^{-\frac{\sigma^2 \|\omega\|^2}{2}} d\omega$, and φ is its characteristic function, then the corresponding ISPD kernel $\varphi(t)$ is the Gaussian $\exp(-\|t\|^2/2\sigma^2)$.

For simplification when computing integral based statistics Bochner's Theorem [2] is employed. This allows for the complex integral expressions commonly produced by the test statistics to be reformulated as U-statistics or von Mises statistics, which can significantly simplify the development of asymptotic theory. The advantage of this reformulation is that the auto-regressive conditional expectation and the auto-regressive conditional probability are both computably easier and also more interpretable, paving the way for a broader potential practical applicability in empirical analysis.

3.4 Application of Mercer's Theorem

(Mercer's) [5] The kernel k has the spectral decomposition:

$$k(x, x') = \sum_{i \in I} \lambda_i e_i(x) e_i(x'), \quad x, x' \in X,$$

where the convergence is absolute and uniform.

- (i) $\{(\lambda_j, e_j)\}$ be pairs of positive eigenvalues and orthonormal eigenfunctions of k .
- (ii) Let X be a compact metric space.
- (iii) $k : X \times X \rightarrow \mathbb{R}$ be a continuous kernel.
- (iv) Let μ be a finite Borel measure with $\text{supp } \mu = X$. Then, for $(e_i)_{i \in I}$ and $(\lambda_i)_{i \in I}$ as above.

In addition the kernel function can be decomposed into its eigenvalues and this process yields a higher analytical power of the proposed tests, facilitated by Mercer's theorem [5]. The fact that the kernel can be expressed in terms of its eigenfunctions and eigenvalues is key in the asymptotic analysis of the corresponding U-statistics. In order to design hypothesis tests with good power and size properties, the limiting behavior of the test statistic under the null and alternative hypotheses is especially informative.

3.5 Testing framework

Model Setup

Actual model

$$Y = M(X) + \varepsilon = \mathbb{E}(Y | X) + \varepsilon$$

where $M_\theta(X)$ is the true underlying model, and ε is the error term.

Assumed model

$$Y = M_{\hat{\theta}}(X) + \varepsilon = \mathbb{E}_{\hat{\theta}}(Y | X) + \hat{\varepsilon}$$

In this section, it is convenient to assume that $\hat{\theta}$ is estimated and known to carry out the following hypothesis testing.

Mathematical Notation

- (i) $Z = (Y, X^\top)^\top$ is a random vector with $\mathcal{Z} \subseteq \mathbb{R}^{1+d}$ and distribution in P_Z
- (ii) X is the explanatory variable which is a random vector taking values in $\mathcal{X} \subseteq \mathbb{R}^d$ with distribution P_X
- (iii) Y represents the real-valued dependent (or response) variable, with $\mathbb{E}|Y| < \infty$
- (iv) $\Theta \in \mathbb{R}^r$, it is a parameter space
- (v) Generalized residual function: $\varepsilon : \mathcal{Z} \times \Theta \rightarrow \mathbb{R}$, and its form is up to $\theta \in \Theta$ where $Y - M_\theta(X) = \varepsilon(Z; \theta)$ and $\varepsilon(X; \theta) = \mathbb{E}(\varepsilon(Z; \theta) | X) = 0$ P_X -a.s.
- (vi) $\mathbb{E}_{\hat{\theta}}(Y | X)$ is called moment restriction models which is the interest that are testing

Hypothesis Testing Problem

Equivalent Hypothesis Testing (1)

Given an i.i.d. sample $\{x_i, y_i; i = 1, \dots, n\}$ drawn from a distribution P_Z , aiming to conduct a specification test:

$$\begin{aligned} H_0 : \varepsilon(X; \hat{\theta}) &= \mathbb{E}(\varepsilon(Z; \hat{\theta}) | X) = 0 \quad (\text{almost surely under the distribution } P_X) \\ H_1 : \varepsilon(X; \hat{\theta}) &= \mathbb{E}(\varepsilon(Z; \hat{\theta}) | X) \neq 0 \quad (\text{almost surely under the distribution } P_X) \end{aligned} \quad \forall \hat{\theta} \in \Theta \quad (1)$$

Where for $\hat{\theta}$, there exists a consistent estimator θ .

Equivalent Hypothesis Testing (2)

Then, the test problem can be conducted in integrated conditional moment (ICM) framework. It is proven that the conditional moment tests in (1) can be transformed into an infinite set of unconditional moment restrictions with the help of weighting function **(STINCHCOMBE and WHITE 1998)** [10] which belongs to a specific class of functions, that is:

$$\varepsilon(X; \hat{\theta}) = \mathbb{E}(\varepsilon(Z; \hat{\theta}) | X) = \mathbb{E}[\varepsilon(Z; \hat{\theta}) h_{\omega}(X)]$$

for all real-valued $h_{\omega}(\cdot)$ such that this expectation exists, and $\forall \omega \in \otimes$, $h \in \mathcal{H}$, where $\mathcal{H}$ is a set of measurable functions on $\mathcal{X}$, then

Here, this thesis takes the weighting function that several ICM tests would follow as Bierens' approaching:

$$h_{\omega}(X) = \exp(i\omega^{\top} \mathbf{X})$$

where:

- (i) $\mathbf{x} \in \mathbb{R}^d$ is the vector of covariates
- (ii) $\omega \in \mathbb{R}^d$ is the frequency parameter
- (iii) i is the imaginary unit

So (1) becomes below:

$$H_0 : \mathbb{E}[\varepsilon(Z; \hat{\theta}) h_{\omega}(X)] = 0 \quad (\text{almost surely under the distribution } P_X) \quad (2)$$

Equivalent Hypothesis Testing (3)

For any $\theta \in \Theta$ that satisfies (2) must also satisfy the following testing (Muandet et.al, 2020) [7].

$$\sup_{h \in \mathcal{H}} \|\mathbb{E}[\varepsilon(Z; \hat{\theta}) h_{\omega}(X)]\|_2^2 = 0 \quad (3)$$

which is called the maximum moment restriction (MMR)

Equivalent Hypothesis Testing (4)

The tested statistic can be viewed as a weighted sum of moment restrictions by Integral-Transform Tests by **STINCHCOMBE and WHITE** in 1998[10]

$$n\mathbb{M}_n^2(\hat{\theta}) = n \iint (R_n(X, t))^\top (R_n(X, t)) d\mu(t) dF(V) \quad (4)$$

where

(i) $t \sim \mathcal{N}(0, 1)$.

(ii) The empirical formulation for $\mathbb{E}[\varepsilon(Z; \theta) h_t(X)]$ is given by:

$$R_n(X, t) = \frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_{\theta_i} h_t(X_i)$$

(iii) Under the null hypothesis, $\sqrt{n}R_n(X, t)$ follow the same distribution as $\varepsilon(Z; \hat{\theta}) h_\omega(X)$.

(iv) $V = \left(X, \varepsilon(Z; \hat{\theta})^\top\right)^\top$

Equivalent Hypothesis Testing (5)

In this testing framework, the ICM test with exponential weighting function $\exp(i\omega^\top X)$ and a uniform integral measure can be stated as in the form of u-statistic using the Gaussian kernel by applying Bocher's Theorem in 3.3.

The Gaussian kernel (also known as Radial Basis Function (RBF) kernel) is defined as:

$$K(\mathbf{x}, \mathbf{y}) = \exp(-\gamma \|\mathbf{x} - \mathbf{y}\|^2)$$

where:

(i) $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ are input vectors

(ii) $\|\cdot\|$ denotes the Euclidean norm

(iii) $\gamma = \frac{1}{2\sigma^2}$ is the kernel parameter ($\sigma > 0$ is the bandwidth)

ICM test statistics in the form of U-statistic:

$$n\mathbb{M}_n^2(\hat{\theta}) = \frac{n}{n(n-1)} \sum_{1 \leq i \neq j \leq n} \hat{\varepsilon}_{\theta_i} \hat{\varepsilon}_{\theta_j} k(x_i, x_j) \quad (5)$$

where $k(x_i, x_j) = \exp\left(-\frac{1}{2} \|x_j - x_i\|^2\right)$, and define: $\hat{h}_2(x_i, x_j) = \hat{\varepsilon}_{\theta_i} \hat{\varepsilon}_{\theta_j} k(x_i, x_j)$

Then restate:

$$n\mathbb{M}_n^2(\hat{\theta}) = \frac{n}{n(n-1)} \sum_{1 \leq i \neq j \leq n} (\hat{h}_2(x_i, x_j) - \mathbb{E}[\hat{h}_2(x_i, x_j)]), \quad \text{with } \mathbb{E}[\hat{h}_2(x_i, x_j)] = 0$$

After applying Mercer's Theorem in 3.4, under the null hypothesis:

$$\begin{aligned}
n\mathbb{M}_n^2(\hat{\theta}) &= \frac{n}{n(n-1)} \sum_{i \neq j} \hat{h}_2(x_i, x_j) = n \left(\frac{1}{n^2} \left[\sum_{i,j} \hat{h}_2(x_i, x_j) \right] - \frac{1}{n^2} \left[\sum_{i=j} \hat{h}_2(x_i, x_j) \right] \right) \\
&= \sum_{l=1}^{\infty} \frac{1}{n} \sum_{i,j} \lambda_l \phi_l(x_i) \phi(x_j) - \sum_{l=1}^{\infty} \frac{1}{n} \sum_{i=1}^n \lambda_l \phi^2(x_i) \\
&= \sum_{l=1}^{\infty} \left[\left(\sum_{i=1}^n \frac{1}{\sqrt{n}} \phi(x_i) \sqrt{\lambda_l} \right) \left(\sum_{j=1}^n \frac{1}{\sqrt{n}} \phi(x_j) \sqrt{\lambda_l} \right) \right] - \sum_{l=1}^{\infty} \frac{1}{n} \sum_{i=1}^n \lambda_l \phi_{(x_i)}^2 \\
&\stackrel{d}{\rightarrow} \sum_{l=1}^{\infty} \lambda_l w_l^2 - \sum_{l=1}^{\infty} \lambda_l \\
&= \sum_{l=1}^{\infty} \lambda_l (w_l^2 - 1)
\end{aligned}$$

Recall the result in 3.2, this also shows that the convergent result of U-statistic, the result above can also be approached below:

$$\begin{aligned}
n\mathbb{M}_n^2(\hat{\theta}) &= \frac{n}{n(n-1)} \sum_{i \neq j} \hat{h}_2(x_i, x_j) \\
&= \frac{2n}{m(m-1)} \cdot \left[\frac{m(m-1)}{n(n-1)} \cdot \left(\sum_{1 \leq i < j \leq n} \hat{h}_2(X_i, X_j) - \mathbb{E}[\hat{h}_2(x_i, x_j)] \right) \right] \\
&= \frac{2n}{m(m-1)} \cdot (\hat{U}_n - \theta) \\
&\stackrel{d}{\rightarrow} \sum_{l=1}^{\infty} \lambda_l (w_l^2 - 1)
\end{aligned}$$

The hypothesis testing can be restated as follows by **Muandet, W. Jitkrittum, and J. Kübler in 2020**[7]:

$$\begin{aligned}
H_1 : n\mathbb{M}_n^2(\hat{\theta}) &\stackrel{d}{\rightarrow} \mathcal{N}(0, 4\sigma_{\theta}^2), \\
H_0 : n\mathbb{M}_n^2(\hat{\theta}) &\stackrel{d}{\rightarrow} \sum_{l=1}^{\infty} \lambda_l (w_l^2 - 1).
\end{aligned}$$

where $\sigma_{\theta}^2 = \text{Var} [\mathbb{E} [h_2(x_i, x_j)]]$

Under H_1 , the test statistic converges in distribution to a normal distribution with mean 0 and variance $4\sigma_{\theta}^2$, where σ_{θ}^2 is the variance of the conditional expectation of the function $\hat{h}_2$.

Under H_0 , the test statistic converges in distribution to a sum involving the eigenvalues λ_l and the terms $(w_l^2 - 1)$.

Equivalent Hypothesis Testing (6)

Following Bierens' approach, thesis focus on the ICM (Integrated Conditional Moment) test statistic expressed in V-statistic form.

The resulting V-statistic form of the ICM test statistic is by Bieren:

$$\begin{aligned}
 nV_n &= n \left(\frac{\hat{\varepsilon}^T}{n} \langle k(x, \cdot), k(x, \cdot) \rangle_{\mathcal{H}} \frac{\hat{\varepsilon}}{n} \right) \\
 &= n \left(\frac{\hat{\varepsilon}_{\theta_i}^T}{n} \cdot K \cdot \frac{\hat{\varepsilon}_{\theta_i}}{n} \right) \\
 &= \frac{n}{n^2} \sum_{1 \leq i, j \leq n} \hat{\varepsilon}_{\theta_i} \hat{\varepsilon}_{\theta_j} k(x_i, x_j) \quad (7)
 \end{aligned}$$

where:

- (i) $\hat{\varepsilon}_{\theta_i}$ are the estimated residuals
- (ii) n is the sample size
- (iii) The double summation averages over all pairs of observations

3.6 Estimation effect

Introduction to Estimation effect

The significance of goodness-of-fit for regression models in particular has to do with the alignment between the specification of the model and the data, as measured by the test statistic. Ideally, this evaluation should be removed from the specific parameter estimation, or the method by which they were derived. This is referred to as the estimation effect, which occurs when variation in the test statistic is due to variation in parameter estimation and not due to the model specification itself. This illustrates the distortion in the goodness-of-fit test caused by the fact that the test statistic directly depends on the estimated parameters rather than on the inherent structure of the model.

Hence, to remove the effect of estimating parameters, also identified in the literature as estimation effect, carrying out a statistical projection of the original kernel, used for composing the test statistic, onto the subspace of residuals. This separates the part of the kernel that is orthogonal to and so whose internal elements that fit to the model parameters, and therefore does not correlate with the estimated parameters.

The fundamental theory of this transformation lies in the kernel matrix, specifically the original kernel matrix, where the kernel matrix contains information on the explanatory variables and specification deviations in the residuals. However, because the goal of the goodness-of-fit statistic is to detect systematic structure in the portion of the data unexplained by the model; any effect due solely to parameter estimation must be removed from the test statistic.

In particular, by applying a projection procedure to the kernel matrix that subtracts the component which is aligned with the explanatory structure of the model. This ensures that only

the variance in the residual space is preserved, making it free of variance due to the estimation process itself. The resulting test statistic therefore describes deviations from the null model specification under the alternative hypothesis and is not affected by deriving from parameter estimation since it is constrained to the space orthogonal to the one spanned by the included regressors.

That in turn makes the test robust not just to the way or precision with which parameters are estimated; it makes it more principled, less arbitrary, in assessing the specification validity of a model. Thus, this projection step is key to separating the intrinsic model structure from the effects of nuisance estimation.

Modified Gaussian Kernel

Recall the generalized residual function, applying the Mean Value Theorem to it:

$$\begin{aligned}\varepsilon(X; \hat{\theta}) &= \varepsilon(X; \theta) + \varepsilon(X; \bar{\theta})'(\theta - \hat{\theta}) \\ &= \varepsilon(X; \theta) + g(\bar{\theta})(\theta - \hat{\theta})\end{aligned}$$

Within current framework, it is not required to explicitly solve for the intermediate parameter value $\bar{\theta}$. According to the Mean Value Theorem this parameter need only satisfy the existence condition, $\bar{\theta} \in (\theta, \hat{\theta})$. The essential idea is that the asymptotic behavior is determined by the gradient of the generalized residual function $\varepsilon(X; \theta)$ at $\bar{\theta}$. Also, considering the estimated parameters as independent variables, this derivative matrix is just equal to the matrix $-X$. Therefore, while $\bar{\theta}$ appears as an intermediate quantity in the theoretical proof, computations when an estimate are available only involve its derivative being a constant which can now be used to calculate all relevant quantities, as the asymptotic variance.

where:

$$\begin{aligned}g(\bar{\theta}) &= \varepsilon(X; \bar{\theta})' \\ &= \varepsilon_X(\bar{\theta})' \\ &= (Y - M_{\bar{\theta}}(X))' \\ &= (Y - X\bar{\theta})' \\ &= -X\end{aligned}$$

Kernel Projection Operation

By **Li and Song (2022)**[6], eliminating the estimation effect, define a new kernel $k_p(x_i, \cdot)$ as:

$$k_p(x_i, \cdot) = k(x_i, \cdot) - g_i(\bar{\theta})(g(\bar{\theta})^T g(\bar{\theta}))^{-1} g(\bar{\theta})^T k(x, \cdot)$$

where:

- (i) $k(x_i, \cdot)$ is the original kernel function valuated at observation x_i
- (ii) $g(\bar{\theta}) = (g_1(\bar{\theta}), \dots, g_n(\bar{\theta}))^\top$ is the gradient matrix of the parametric valuated at the parameter $\bar{\theta}$
- (iii) $g_i(\bar{\theta})(g(\bar{\theta})^\top g(\bar{\theta}))^{-1}g(\bar{\theta})^\top$ is the projection matrix onto the space spanned by $g(\bar{\theta})$ based on the i -th observation.

The operation $k_p(x_i, \cdot)$ projects $k(x_i, \cdot)$ onto the orthogonal complement of the subspace generated by $g(\bar{\theta})$, removing linear dependence on the parametric component and analyze further under the nonparametric only.

Original Test Statistic

The test statistic is:

$$\begin{aligned} nV_n &= n \left(\frac{\hat{\varepsilon}^\top}{n} \langle k(x, \cdot), k(x, \cdot) \rangle_{\mathcal{H}} \frac{\hat{\varepsilon}}{n} \right) \\ &= \frac{1}{n} \langle \hat{\varepsilon} k(x, \cdot), \hat{\varepsilon} k(x, \cdot) \rangle_{\mathcal{H}} \end{aligned}$$

Effect of Modified Kernel

Using the new kernel $k_p(x, \cdot)$, showing the estimated parameter has no effect:

$$\begin{aligned} \hat{\varepsilon} k_p(x, \cdot) &= \varepsilon(X; \hat{\theta}) k_p(x, \cdot) \\ &= (\varepsilon(X; \theta) + g(\bar{\theta})(\theta - \hat{\theta}))^\top k_p(x, \cdot) \\ &= \varepsilon(X; \theta) k_p(x, \cdot) + g(\bar{\theta})(\theta - \hat{\theta})^\top k_p(x, \cdot) \\ &= \varepsilon(X; \theta) k_p(x, \cdot) + g(\bar{\theta})(\theta - \hat{\theta})^\top (k(x, \cdot) - g(\bar{\theta})(g(\bar{\theta})^\top g(\bar{\theta}))^{-1}g(\bar{\theta})^\top k(x, \cdot)) \\ &= \varepsilon(X; \theta) k_p(x, \cdot) + ((\theta - \hat{\theta}))g(\bar{\theta})^\top k(x, \cdot) \\ &\quad - (\theta - \hat{\theta})(g(\bar{\theta})^\top g(\bar{\theta}))(g(\bar{\theta})^\top g(\bar{\theta}))^{-1}g(\bar{\theta})^\top k(x, \cdot) \\ &= \varepsilon(X; \theta) k_p(x, \cdot) + ((\theta - \hat{\theta}))g(\bar{\theta})^\top k(x, \cdot) \\ &\quad - (\theta - \hat{\theta}) I g(\bar{\theta})^\top k(x, \cdot) \\ &= \varepsilon(X; \theta) k_p(x, \cdot) + ((\theta - \hat{\theta}))g(\bar{\theta})^\top k(x, \cdot) \\ &\quad - (\theta - \hat{\theta})g(\bar{\theta})^\top k(x, \cdot) \\ &= \varepsilon(X; \theta) k_p(x, \cdot) \\ &= \varepsilon k_p(x, \cdot) \end{aligned}$$

The key observation is that pre-orthogonalized kernel function $k(x, \cdot)$ has already project the part of parametric gradient $g(\bar{\theta})$. This orthogonality implies that all terms with a bias in the parameters vanish in the expansion. The end result shows that after such an orthogonalization, the modified kernel function estimate $\hat{\varepsilon} k_p(x, \cdot)$ is equal to the true error $\varepsilon k_p(x, \cdot)$ while that for

nonparametric estimation is made immune to the errors in parametric estimation. This approach represents well the principle in semiparametric modelling in that separates parametric part and nonparametric part and leads to the theoretical justifications for subsequent statistical inference.

Modified Test Statistic

The new ICM test statistic with the modified kernel is:

$$\begin{aligned} nV_{n,k_p} &= \frac{1}{n} \langle \hat{\varepsilon}k_p(x, \cdot), \hat{\varepsilon}k_p(x, \cdot) \rangle \\ &= \frac{1}{n} \langle \varepsilon k_p(x, \cdot), \varepsilon k_p(x, \cdot) \rangle \\ &= \frac{1}{n} \sum_{1 \leq i, j \leq n} \hat{\varepsilon}_{\theta i} \hat{\varepsilon}_{\theta j} k_p(x_i, x_j) \end{aligned}$$

Kernel Matrix Computation

The modified kernel $k_p(x_i, x_j)$ is computed as:

$$\begin{aligned} k_p(x_i, x_j) &= \langle k_p(x_i, \cdot), k_p(x_j, \cdot) \rangle \\ &= (k(x_i, \cdot) - g_i(\bar{\theta})(g(\bar{\theta})^T g(\bar{\theta}))^{-1} g(\bar{\theta})^T k(x, \cdot))^T \\ &\quad (k(x_j, \cdot) - g_j(\bar{\theta})(g(\bar{\theta})^T g(\bar{\theta}))^{-1} g(\bar{\theta})^T k(x, \cdot)) \\ &= \text{Term 1: } k(x_i, x_j) \\ &\quad + \text{Term 2: } (g_j(\bar{\theta})^T G g(\bar{\theta})^T) k(x_i, x) \\ &\quad + \text{Term 3: } k(x_j, x)^T (g_i(\bar{\theta})^T G g(\bar{\theta})^T)^T \\ &\quad + \text{Term 4: } g_i(\bar{\theta})^T G g(\bar{\theta})^T K g(\bar{\theta})^T G g_j(\bar{\theta})^T \end{aligned}$$

Where $g(\bar{\theta}) = -X$ as mentioned above, and noting that $(g(\bar{\theta})^T g(\bar{\theta}))^{-1} = G$, and $G^T = G$.

In particular, this expansion is composed of four terms that: the first term integrates the metric properties of the original kernel, whereas the last three terms systematically remove the bias of the parameter estimates. Mathematically, this construction enforces the orthogonality $\langle k_p(x_i, \cdot), g_i(\bar{\theta}) \rangle = 0$, and in this way the nonparametric component will be separated from the parametric dimension. From a computational point of view, it can be achieved efficiently using block decomposition of the kernel matrix. In principle, it combines the power of nonlinear model fitting available in kernel methods with the asymptotic efficiency properties characteristic of semiparametric estimation.

Final Kernel Matrix Expression

The orthogonalized kernel matrix K_p has the following matrix representation:

$$K_p = K + M + M^\top + \Gamma K \Gamma^\top$$

where the components are defined as:

- (i) Original kernel matrix: $K \in \mathbb{R}^{n \times n}$
- (ii) cross product term: $M = \Gamma K$ with matrix $\Gamma = g(\bar{\theta}) G g(\bar{\theta})^\top$
- (iii) Quadratic term: $\Gamma K \Gamma^\top$

This matrix factorization provides three computational benefits: 1) The matrix product representation allows the use of well-optimized linear algebra operations, leading to significantly improved simulation throughput for large data sets; 2) The symmetric representation $M + M^\top$ ensures numerical stability by maintaining matrix condition numbers; and 3) The four-term decomposition can be readily parallelized, as each of the components can be computed independently before they are recombined.

Final Test Statistic Form

The ICM test statistic in V-statistic form with modified kernel is:

$$\begin{aligned} nV_{n,k_p} &= \frac{1}{n} \sum_{1 \leq i, j \leq n} \hat{\varepsilon}_{\theta i} \hat{\varepsilon}_{\theta j} k_p(x_i, x_j) \\ &= \frac{\varepsilon(X; \hat{\theta})^\top K_p \varepsilon(X; \hat{\theta})}{n} \\ &= \frac{\varepsilon(X; \hat{\theta})^\top (K + M + M^\top + \Gamma K \Gamma^\top) \varepsilon(X; \hat{\theta})}{n} \end{aligned}$$

Notation Definitions:

- (i) $\mathbf{Y}$: An $n \times 1$ vector (response variable).
- (ii) $\mathbf{X}$: An $n \times q$ matrix (design matrix with n samples and q variables).
- (iii) $\boldsymbol{\theta}$: A $1 \times q$ row vector (parameter vector).
- (iv) $\boldsymbol{\varepsilon}$: An $n \times 1$ vector (all residual related terms).
- (v) $\mathbf{K}$: An $n \times n$ Gaussian kernel matrix, where $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.
- (vi) $\mathbf{K}_p$: An $n \times n$ modified Gaussian kernel matrix, where $K_{p,ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.
- (vii) $k(\mathbf{x}_i, \mathbf{x})$ or $k_p(\mathbf{x}_i, \mathbf{x})$: The i -th column of $\mathbf{K}$ or $\mathbf{K}_p$ ($n \times 1$ vector).
- (viii) $g(\bar{\theta}) = -\mathbf{X}$: An $n \times q$ matrix, where $g_i(\bar{\theta})$ (the i -th row) is a $1 \times q$ row vector.

3.7 Multiplier bootstrap

This thesis proposes the use of the multiplier bootstrap method in order to obtain the test statistic critical value, which has a significant computational advantage compared to other bootstrap techniques. First, unlike the other bootstrap techniques, it does not need model parameters to be re-estimated at each resampling iteration. Instead, this procedure induces random perturbations directly on the residuals from the original sample and does not require pseudo-sample generation nor model re-estimation. This can greatly reduce the computational load, especially in large sample cases.

Second, deriving the test statistic by projecting the kernel matrix onto the subspace of residual errors, so that the effects from parameter estimation are removed. Thus, the procedure is also not dependent upon either the parameter estimators having an asymptotic linear representation, as is almost always required in conventional theoretical approaches. This makes the multiplier bootstrap highly versatile across a vast class of models, and consequently ensures both theoretical rigor and practical relevance.

Algorithm

Notation Definitions:

- (i) modified kernel matrix K_p .
- (ii) residual: $\varepsilon(X; \hat{\theta})$
- (iii) Number of bootstrap samples $B \in \mathbb{N}$
- (iv) Confidence level $\alpha \in (0, 1)$

1. Preprocessing:

$$\varepsilon(X; \hat{\theta}) = (\epsilon_1, \epsilon_2, \dots, \epsilon_n)^T$$

2. Multiplier Bootstrap:

For $b = 1$ to B do:

(a) Generate random weights:

$$W^{(b)} = (W_1^{(b)}, \dots, W_n^{(b)})^T, \quad W_i^{(b)} \sim \mathcal{N}(0, 1)$$

(b) Compute weighted errors:

$$\epsilon^{(b)} = W^{(b)} \odot \varepsilon(X; \hat{\theta})$$

(c) Calculate quadratic form:

$$V_b = \frac{1}{n} \epsilon^{(b)T} K_p \epsilon^{(b)}$$

3. Critical Value Calculation:

$$\hat{c}_\alpha = \text{Quantile}(\{V_1, \dots, V_B\}, \alpha)$$

The algorithm estimates the distribution of:

$$nV_{n,k_p} = \frac{\varepsilon(X; \hat{\theta})^T K_p \varepsilon(X; \hat{\theta})}{n}$$

using the *multiplier bootstrap* method, which preserves the structure of $\varepsilon(X; \hat{\theta})$ through random weighting

It is, informally, generating the residual vector and adding an independent symmetric randomness to it then computing a statistic on it and doing it many times in order to obtain the distribution of the statistic and pick up its critical value from the Confidence Level.

In probability and statistics the equivalent to cross-validation applies to testing the residuals of a model through resampling using symmetric randomization. Specifically, applying the random symmetric perturbation with mean of zero and variance zero *i.i.d*, and referring to the unperturbed pseudo-residuals after forming these perturbed pseudo-residuals. Second, the test statistic is re-computed using these pseudo-residuals. This is repeated B times to obtain the empirical sampling distribution of the test statistic under the null hypothesis. Finally, at a fixed pre-specified level α (e.g. $\alpha = 0.05$ or 0.01), one extracts from this empirical distribution the corresponding critical value based on the chosen percentile (e.g. 95-th or 99-th percentile) of this empirical distribution. This non-parametric approach preserves the symmetry of the analyzed dataset, avoiding parametric constraints, while re-sampling preserves the robustness of statistical inferences –a distinct advantage when hypothesis generation suffers from model misspecification, small sample sizes, or both.

Chapter 4

Simulation

4.1 Parameter Settings

Category	Symbol/Variable	Value
Data Generation (DGP)	X	$n = 1000$ (sample size)
	X	$k = 5$ (number of convirate variables)
	β	$[1.0, 2.5, 3.0, 3.5, 4.0, 4.5]^T$ (coefficents)
	σ	$\sigma = 1$ (variance of npise)
Kernel Function	sigma kernel	$\sigma_{\text{kernel}} = 2$
Bootstrap Test	times:N	500(Number of repeats)
	confidence_level	90,95,99th percentile
Rejection Rate	repeats:R	500 (Number of decision making)
Alternative Hypotheses	H1_norm	$\ X\ _2$
	H1_sin	$\sin(\ X\ _2)$
	H1_cos	$\cos(\ X\ _2)$
	H1_sincos	$\sin(\ X\ _2) + \cos(\ X\ _2)$
	H1_interaction	$X_1^2 + X_1 X_2$

Table 4.1: Summary of Key Parameters

The experimental design is formed by combining the parameters of four main components: data generation part, kernel method part and bootstrap testing and nonlinear Parttern.

(1)Data Generation: The large sample size (1000) to guaranties that all the properties are large sample properties, to avoid random fluctuations. Linear coefficients define baseline relationships, and $\sigma = 1$ modulates noise levels in order to mimic standard normal uncertainty.

(2)Kernel Function: Local sensitivity is controlled by the RBF kernel bandwidth. The former represents short range effects, such as the detection of high-frequency nonlinearities (e.g., 0.5) and the latter represents the long-ranged effects, such as the capture of global patterns (e.g., 5).

(3) Bootstrap Testing: There is a joint determination of the number of resampling iterations and confidence level (90,95,99th percentile) for critical value estimation.

(4) Nonlinear Patterns: In five alternative hypothesis (norm, trigonometric, interaction 5 terms) are exhaustively considered to evaluate the ability of detection of the kernel method and testing framework.

4.2 Model Specification and Hypothesis Testing setting

Consider an observable dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^p$ are control variables and $y_i \in \mathbb{R}$ is the response variable. This thesis formulates the following statistical hypothesis test to examine the specification of the generating data:

Null Hypothesis (H_0): Linear Regression Model

The data can be fully captured by a linear regression model, such that there exists a true parameter vector $\boldsymbol{\beta} \in \mathbb{R}^p$ and residual variance $\sigma^2 > 0$ satisfying:

$$y_i = \mathbf{x}_i \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n$$

where:

- (i) $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top$ is the design matrix
- (ii) Error terms ε_i are normally distributed with $\mathbb{E}[\varepsilon_i | \mathbf{X}] = 0$ and $\text{Var}(\varepsilon_i | \mathbf{X}) = \sigma^2$

Alternative Hypothesis (H_1): Nonlinear Regression Model

The data shows significant nonlinear dependence that cannot be adequately captured by a linear model. Specifically, there exists at least one nonlinear measurable function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ such that:

$$y_i = f(\mathbf{x}_i; \boldsymbol{\theta}) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

where:

- (i) $f(\cdot; \boldsymbol{\theta})$ represents the nonlinear structure with parameters $\boldsymbol{\theta} \in \mathbb{R}^k$
- (ii) No linear mapping $\boldsymbol{\beta}$ exists such that $f(\mathbf{x}; \boldsymbol{\theta}) = \mathbf{x}^\top \boldsymbol{\beta}$ for all $\mathbf{x}$ (i.e., f cannot reduce to linear form)

4.3 Data Generating Process

Null Hypothesis Data Generation

- Independent variables:

$$\mathbf{X} \in \mathbb{R}^{n \times 5}, \quad X_{ij} \sim \mathcal{N}(0, 1) \quad \forall i = 1, \dots, n; j = 1, \dots, 5$$

- Design matrix with intercept:

$$\tilde{X} = [\mathbf{1}_n \mid X] \in \mathbb{R}^{n \times 6}$$

- Response variable:

$$Y = \tilde{X}\beta + \varepsilon, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Alternative Hypothesis Data Generation

(a) L2-Norm Nonlinearity (generate_data_H1_norm)

$$Y_{H1} = \tilde{X}\beta + \underbrace{\|X\|_2}_{\text{nonlinear term}} + \varepsilon, \quad \|X_i\|_2 = \sqrt{\sum_{j=1}^5 X_{ij}^2}$$

This term is the integrated multi-dimensional features that indicates geometric magnitude of the feature vector in $\mathbb{R}^5$.

(b) Cosine Nonlinearity (generate_data_H1_cos)

$$Y_{H1} = \tilde{X}\beta + \cos(\|X\|_2) + \varepsilon$$

The $\cos(\|X\|_2)$ term yields smooth oscillatory patterns that depend on the norm of X , introducing periodic relationships not considered in linear models.

(c) Sine Nonlinearity (generate_data_H1_sin)

$$Y_{H1} = \tilde{X}\beta + \sin(\|X\|_2) + \varepsilon$$

The $\sin(\|X\|_2)$ term introduces nonlinear corrections that have a $\frac{\pi}{2}$ phase shift with respect to the cosine version, probing sensitivity to different periodic structures.

(d) Mixed Sinusoidal Terms (generate_data_H1_sincos)

$$Y_{H1} = \tilde{X}\beta + \sin(\|X\|_2) + \cos(\|X\|_2) + \varepsilon$$

The hybrid structure enhances nonlinear mechanism by superposing of two counteractive modes with interference pattern.

(e) Interaction & Quadratic Terms (generate_data_H1_interaction)

$$Y_{H1} = \tilde{X}\beta + \underbrace{X_{.,1} \odot X_{.,1} + X_{.,1} \odot X_{.,2}}_{\text{nonlinear terms}} + \varepsilon$$

Here: $X_{.,1} \odot X_{.,1}$ is a parabolic deviation for the first feature, $X_{.,1} \odot X_{.,2}$ are used to represent cooperative effects between first and second order features. This device tests model quality in polynomial and multiplicative non-linearities.

Notation Definitions:

- (i) Y : Response vector ($n \times 1$)
- (ii) X : Independent Vars ($n \times 5$)
- (iii) $\tilde{X}$: Design matrix ($n \times 6$)
- (iv) $\varepsilon_{\text{real}}$: Error vector ($n \times 1$)
- (v) $\beta \in \mathbb{R}^6$: Coefficients (including intercept β_0)
- (vi) σ : Noise standard deviation
- (vii) n : Sample size (row dimension)

Visualization of Data Flow:

$$\begin{array}{ccc}
 \text{Independent Vars} & \text{Design Matrix} & \text{Response} \\
 X \in \mathbb{R}^{n \times 5} & \tilde{X} \in \mathbb{R}^{n \times 6} & Y \in \mathbb{R}^{n \times 1} \\
 \downarrow & \downarrow & \downarrow \\
 \begin{bmatrix} X_{11} & \cdots & X_{15} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{n5} \end{bmatrix} & \begin{bmatrix} 1 & X_{11} & \cdots & X_{15} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{n5} \end{bmatrix} & \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}
 \end{array}$$

In the procedure of generating data, there is a prespecified beta that is a known, fixed parameter. This parameter is the underlying coefficient vector for generating the linear part of the synthetic data. X and matrix $\tilde{X}$ (which includes a column of ones) of the independent variables are then generated on the basis of standardized probabilistic distributions. Added to such a set of linear predictors are two deterministic terms, the first being an independent and identically distributed random noise sampled from some specified probability distribution, and the second being a nonlinear term which is the result of explicit transformations of X . The final observed outcome variables are the product of all these variables.

Note that even though the parameter β is considered a known constant in the data synthesis, all further hypothesis testing procedures are based on testing a null hypothesis that the parameter β is unknown, and that it needs to be estimated from the observed data using standard available statistical tools. The analysis centers on assessing to what extent the proposed adjusted kernel matrix successfully removes the biases due to the estimation uncertainty of beta. This study also provides more information about the probabilistic reliability of the resulting test statistics to correctly identify the true model structure—the differentiation between the null (linearity) and alternative hypotheses (nonlinearity)—at finite sample sizes. Parameter uncertainty should be incorporated into inferential validity given a particular model.

4.4 Measurement metric: Rejection Rate

Parameter Estimation via OLS

Under the null, the OLS method estimates regression coefficients $\hat{\beta}$. Residuals $\hat{\varepsilon}$ capture deviations from the linear model.

$$\begin{aligned}\hat{\beta} &= (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T Y \\ \hat{\varepsilon} &= Y - \tilde{X} \hat{\beta}\end{aligned}$$

Modified Kernel Matrix estimation effect

An RBF kernel K measures pairwise similarities. The modified kernel K_p removes linear dependence on the parameter estimation.

$$K_p = K + M + M^\top + \Gamma K \Gamma^\top$$

where the components are defined as

- (i) $(X^T X)^{-1} = G$
- (ii) Original kernel matrix: $K \in \mathbb{R}^{n \times n}$
- (iii) cross product term: $M = \Gamma K$ with matrix $\Gamma = X G g X^\top$
- (iv) Quadratic term: $\Gamma K \Gamma^\top$

where:

$$K_{ij} = \exp(-\gamma \|\tilde{X}_i - \tilde{X}_j\|^2), \quad \gamma = \frac{1}{2\sigma_{\text{kernel}}^2}$$

Computing test statistics

The ICM test statistics nV_{n,k_p} measures unexplained nonlinear patterns in residuals.

$$nV_{n,k_p} = \frac{1}{n} \hat{\varepsilon}^T K_p \hat{\varepsilon}$$

Multiplier Bootstrap for Critical Value Approximation

Gaussian weights perturb residuals to compute bootstrap statistics V_b .

$$W \sim \mathcal{N}(0, I_n), \quad \tilde{\varepsilon}_b = W \odot \hat{\varepsilon}$$

$$nV_{n,k_p} = \frac{1}{n} \tilde{\varepsilon}_b^T K_p \tilde{\varepsilon}_b$$

Repeat this procedure for N times:

$$c_\alpha = \text{Quantile}(\{nV_{n,k_p}\}, 1 - \alpha)$$

Rejection Decision and Empirical Rejection Rate Calculation

Rejection rate is computed by averaging results over R experiments

$$\text{Reject} \iff V > c_\alpha$$

Rate under H_0 close to α indicates test size (Type I error), and higher rates under H_1 show test power (Type II error).

$$\text{Rejection Rate} = \frac{1}{R} \sum_{r=1}^R \mathbb{I}(V^{(r)} > c_\alpha^{(r)})$$

In the context of hypothesis testing, the rate of rejections is a key performance metric for assessing the test. When the data really fit the null hypothesis H_0 , the rejection rate is the significance level (test size), which also measures Type I error rate α , which is the probability of wrongfully rejecting H_0 when it is true. For example, if $\alpha = 0.05$, the rejection probability under H_0 should, in principle, be close to 5%. Empirical testing in simulations (e.g., simulating R datasets under H_0 is essential to calibrate the test: a rejection rate close to α is evidence that the test works, a substantially higher rate means that are at an increased risk of false positives, lower-rate – that are overly conservative.) This procedure verifies the capacity of the test to handle type-I errors at the null hypothesis H_0 .

When within the alternative hypothesis H_1 , the proportion of time that the test rejects is the statistical power (test power), the probability of rejecting H_0 given that H_1 is true (i.e., $1-\beta$), where β is the Type II error rate). In the current conceptual space, the rejection rate needs to be much higher than α (preferably close to 1, depending on the value of the effect size, the sample size, the noise level and α). A rejection rate significantly less than α indicates inadequate power and the need for modifications (e.g., larger sample sizes). Comparing the rejection rate under H_0 with the rejection rate under H_1 allows to scrutinize and balance the tradeoff between type I error control and sensitivity to true effects, in order to obtain a test that is robust against false claims and powerful to detect real signals.

4.5 Results

Based on the specified linear model of interest, this thesis investigates the distribution behavior of V-statistic to different number of multiplier bootstrap replication N . The simulation results (Figs. 1-8) illustrate that, for $N = 30, 50$ or 100 , the empirical distributions deviate significantly from the asymptotic distribution for $N = 3000$ (which considered as our population). In particular, it is clear that the shape of the distribution is on the verge of converging and stabilizing and begins to show any kind of acceptable convergence with no longer N approximately 200.

Given these results, and to guarantee strong statistical proof, to make a conservative choice, and use $N=500$ in our analysis. This option guarantees enough buffer for computation efficiency and accurate estimation of the critical value.

The tables and histograms below shows that the rejection rates of both H_0 and H_1 are consistent with the theory when $n = 1000$. Nevertheless, for 200 samples, the rejection rate of generated experimental data differ considerably from their expected values. It can be shown that the bandwidth of gaussian kernel also plays an important role.

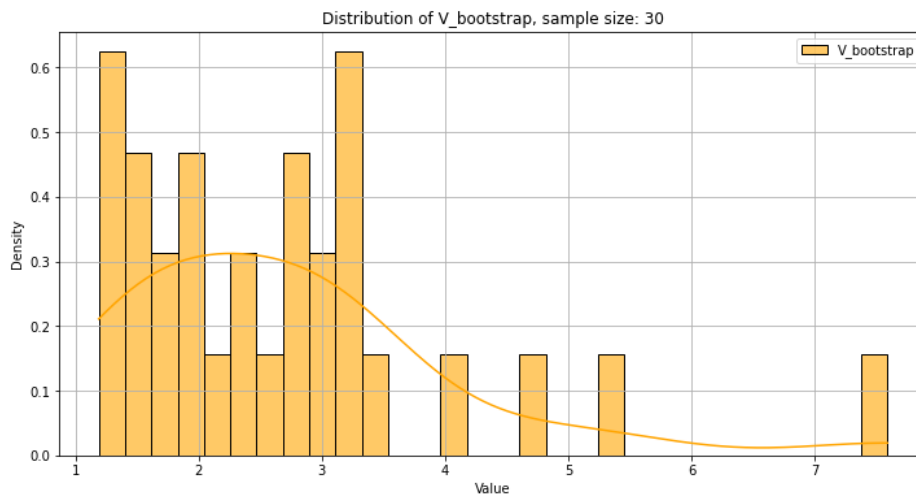


Figure 4.1: $N=30$, distribution of V_s statistics

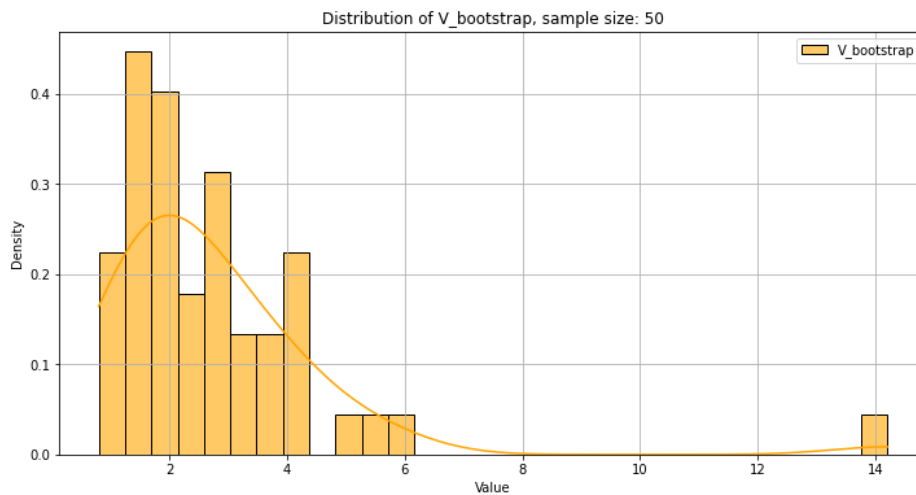
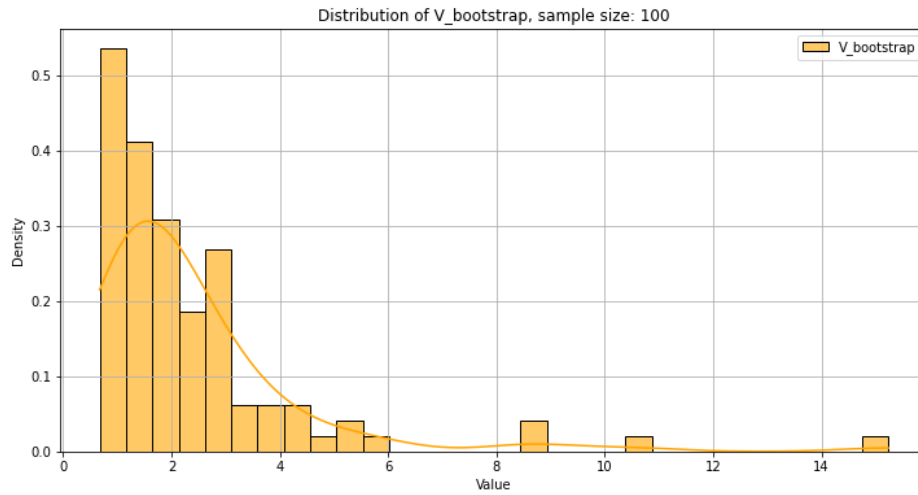
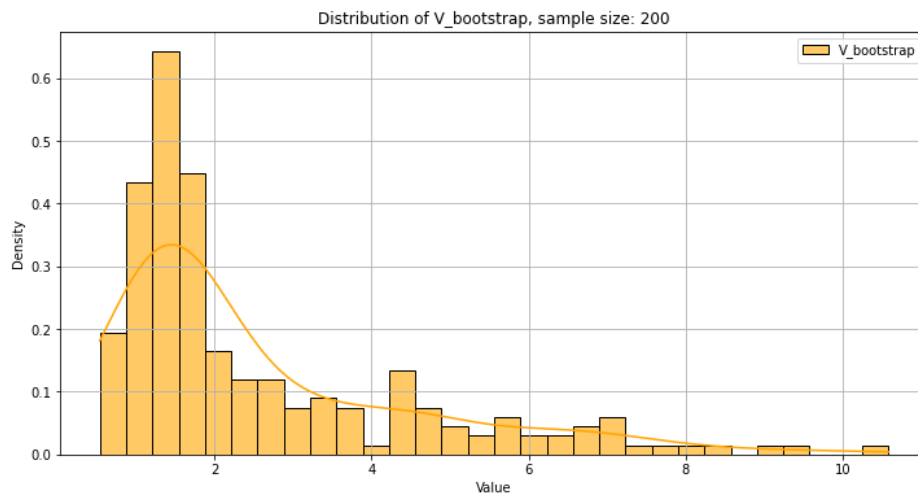
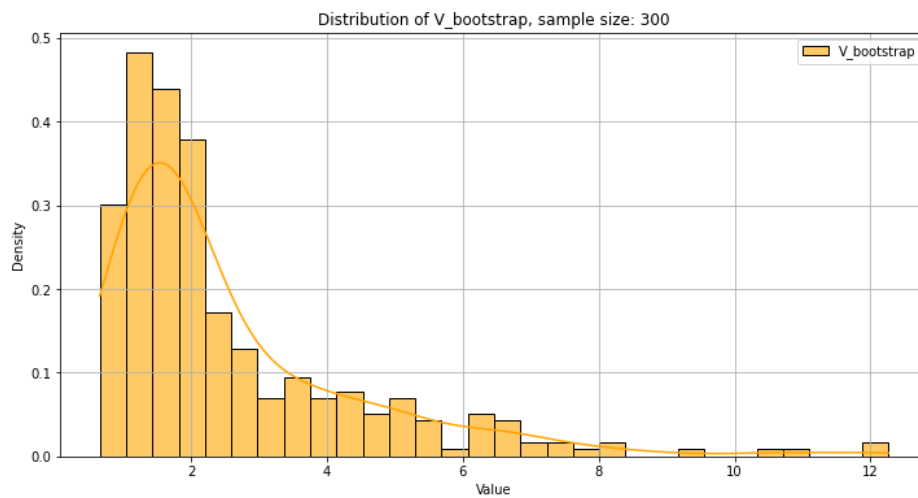
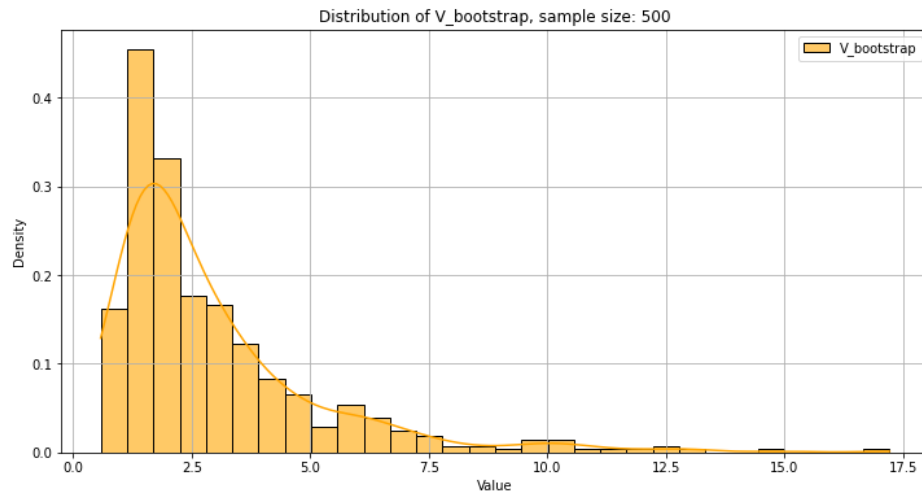
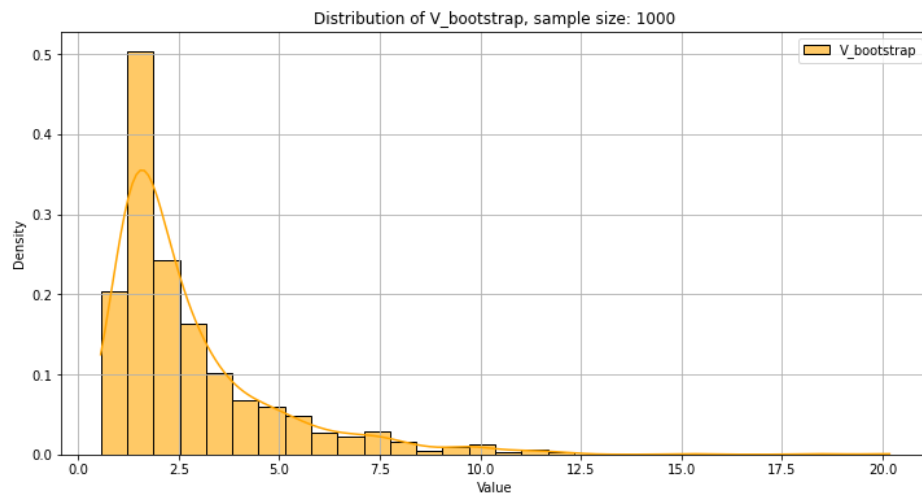
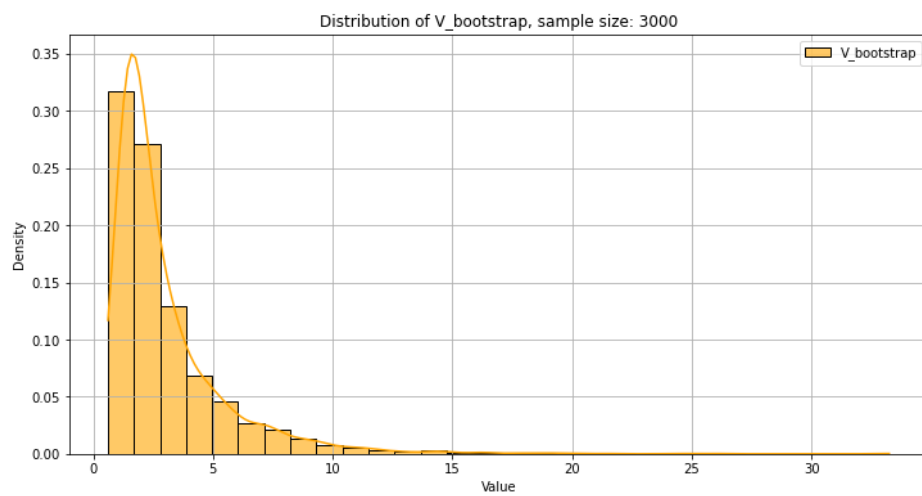


Figure 4.2: $N=50$, distribution of V_s statistics

Figure 4.3: $N=100$, distribution of V_s statisticsFigure 4.4: $N=200$, distribution of V_s statisticsFigure 4.5: $N=300$, distribution of V_s statistics

Figure 4.6: $N=500$, distribution of V_s statisticsFigure 4.7: $N=1000$, distribution of V_s statisticsFigure 4.8: $N=3000$, distribution of V_s statistics

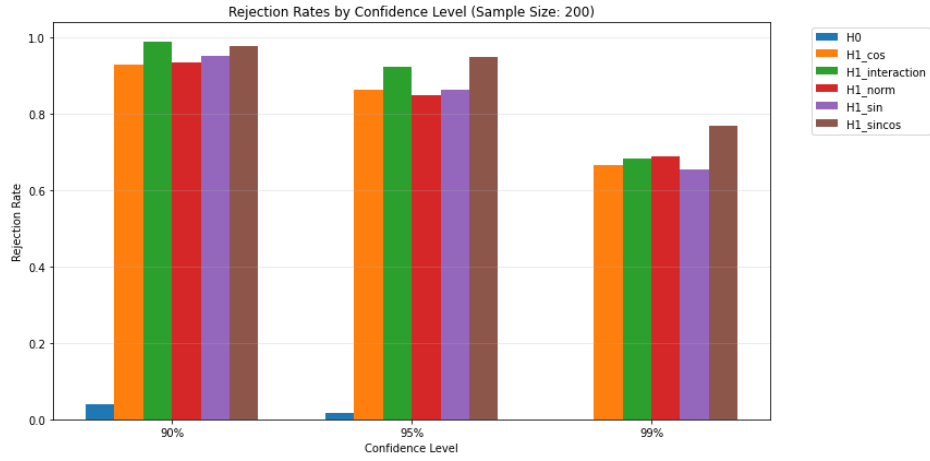


Figure 4.9: Rejection rates under different confidence levels for each DGP (sample size:200)

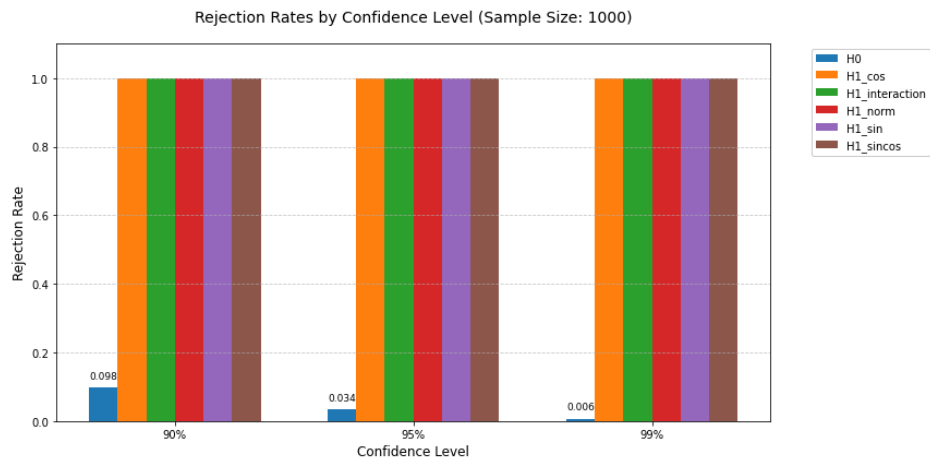
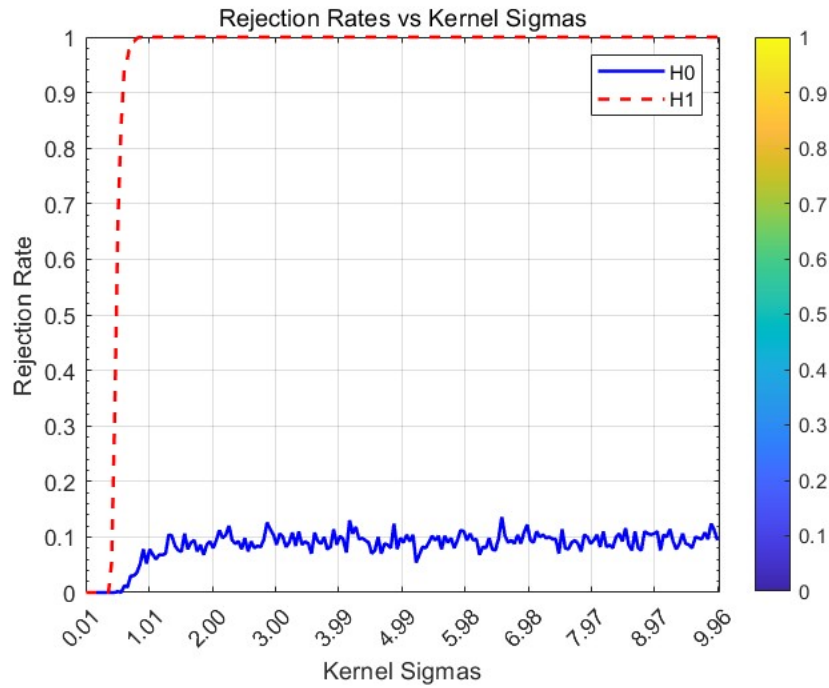


Figure 4.10: Rejection rates under different confidence levels for each DGP (sample size:1000)

Figure 4.11: kernel.vs.rejection rate under H_0

Confidence Level	H0	H1_cos	H1_interaction	H1_norm	H1_sin	H1_sincos
90%	0.04	0.93	0.99	0.936	0.952	0.978
95%	0.016	0.862	0.924	0.848	0.862	0.95
99%	0	0.666	0.682	0.689	0.654	0.77

Table 4.2: Rejection rates under different confidence levels for each DGP (sample size:200)

Confidence Level	H0	H1_cos	H1_interaction	H1_norm	H1_sin	H1_sincos
90%	0.098	1.0	1.0	1.0	1.0	1.0
95%	0.034	1.0	1.0	1.0	1.0	1.0
99%	0.006	1.0	1.0	1.0	1.0	1.0

Table 4.3: Rejection rates under different confidence levels for each DGP (sample size:1000)

Chapter 5

Discussion

Test power and test size

Rejection rates under H_0 (at confidence level 0.90, 0.95 and 0.99), when the sample size of the DGP is 200, are 0.04, 0.016 and 0. This finding implies that, while the confidence level is increasing, the rejection rate of H_0 is decreasing, as theoretically anticipated. Nevertheless, the observed values (0.04, 0.016, 0) are far from the ideal theoretical values (0.1, 0.05, 0.01), indicating that the present test power still not reached the desired accuracy and further increases of sample size should be considered for increasing the testing accuracy.

Rejection rates are uniformly high across the average of all five alternatives H_1 , but again decrease with higher confidence levels. This shows that the testing procedure is capable of strong statistical power in testing against the null hypothesis. Additionally, with higher confidence levels a higher significance threshold is used, and the decrease in rejection rates can be interpreted as an indicator that thresholds for robustness of the results against a higher level of confidence are higher as well.

The rejection rates under H_0 follow the size properties of the test, and the deviations from the theoretical ones indicate that sample size calibration could be improved. On the other hand, the persistently high rejection rates under H_1 justify the test in terms of the statistical sense: the current method discriminates well the null hypothesis from the alternative. These results are consistent with the general soundness of the testing scheme, though they suggests that further improvement will be required to maintain high statistical power at stricter confidence levels.

The empirical rejection rates under the null H_0 at confidence level 0.90, 0.95, and 0.99 are 0.098, 0.034, 0.006 when the sample size from the DGP is 1000. Remarkably, these rejection rates not only fall as the CL is tightened—as is theoretically expected—but also closely match their target levels (0.1, 0.05, and 0.01), indicating a better calibration than at the lower sample sizes. Crucially, under all five alternatives models H_1 , the successive significance levels all reach the maximum of 1.00, suggesting that the test has superb discriminatory power, even under the strictest confidence scenario. The results support the conclusion that a sample size of 1000 is large enough to provide a controlled test size (Type I error) and near-optimal test power (Type

II error). In addition, comparing sample sizes of 200 and 1000 shows clear improvement of empirical versus theoretical rejection rates indicating the necessity of larger number of samples to make testing framework valid and reliable.

The impact of Gaussian kernel bandwidth

The contribution of the bandwidth of the Gaussian kernel to infer statistical test statistics is mainly in trading off smoothness to sensitivity. A high bandwidth causes overt smoothness by the kernel function of the data. Although it may help in attenuating the noise corruption and in minimizing the variance of the estimate, this smoothing could hide substructure of the signal and make test statistic less sensitive to even small discrepancies, and thus weaken the testing power. A narrower bandwidth, on the other hand, results in sharper detection of local variations, which is more sensitive, but is more vulnerable to random fluctuations, increasing estimation variability and the possibility of inducing spurious peaks causing false positive error.

The simulation part aims to study the effects of Gaussian kernel bandwidth on the rejection rates of the residual-based kernel test under null hypothesis H_0 and the alternative hypothesis H_1 . It is found that the test size is well controlled by the nominal significance level of 0.1 with respect to all values of bandwidth(except around 0), suggesting that the method can have good power in controlling Type I errors. Under H_1 , in the region of bandwidth almost equal to zero value, test power increases very fast approaching one as the bandwidth tends to be one that remains high as bandwidth is increased.

In comparison to the initial theoretical expectation, where test power decreases if the bandwidth is too large, observing that the power is maintained when the bandwidth varies. This may because, first, the nonlinear part of the data varies globally a lot, which makes the test able to capture departure from linearity even if the kernel is getting flatter; second, the sample size of 1000 is large enough that the test statistic has a high power against large bandwidth; and third, although using a larger bandwidth will smooth the kernel matrix, the signal is much larger than the noise, and the test can still separate H_0 and H_1 .

Chapter 6

Conclusion

The paper is motivated in the context of Bierens' s Integrated Conditional Moment (ICM) testing approach. Instead of directly computing the involved complex integral transforms in and over which the test statistics are traditionally computed, thesis uses a test statistic using a reproducing kernel Hilbert space (RKHS) approach, with the Gaussian kernel. It achieves this by bypassing the theoretical and computational difficulties related to calculation of integrals in the analytic solution and allows the test to classify general types of model misspecification, especially nonlinear aspects in simulation part, without a prior specification of the shape of the misspecification.

After the definition of the Gaussian-kernel- based test statistic, this thesis investigates the asymptotic behavior of this test statistic and the corresponding V-statistic. But there is a serious issue as the kernel matrix constructed directly from the design matrix X , is correlated with the regressors. This correlation reflects the parts explained by the linear part of the model—something that instead is not of interest in a testing framework since the aim is to test the misspecification.

To deal with this, an modified kernel matrix is introduced. This modified kernel is obtained by orthogonally projecting the original Gaussian kernel matrix onto the space spanned by the predictors, hence asymptotically removing the estimation effect. This modification makes the test statistic responsive only to the components of variability in the data explained by the linear part and significantly enhances the validity and power of the testing procedure, especially under null alternatives.

In this empirical study, a V- statistic under the modified kernel framework is used as the test statistic. The Multiplier Bootstrap technique is employed in order to estimate critical values for hypothesis testing. This approach has the computational advantage of sampling for repeated samples from the null distribution, yet provides consistent estimators for the finite-sample null-based distribution of the test statistic. By extensive simulation evidence, the finite-sample performance of the proposed method is examined over various experimental settings, which verifies the theoretical analyses under the asymptotic regime.

One important empirical observation is the influence of sample size on the performance of the test. Larger sample sizes not only make the asymptotic distribution better approximated but also place heavier demands on tuning parameter selection, such as the number of bootstrap replications and the kernel bandwidth. If the high confidence was considered, cautious should be taken when choosing the parameters to ensure the validity of the inference.

In addition, the effects of the smoothing bandwidth in the Gaussian kernel on the test size and test power are systematically investigated. For the test size, experiments confirm the theoretical expectation: as the bandwidth tends to be zero, the kernel becomes too localized to catch the global structure of model misspecifications and the rejection rate falls down. As bandwidth rises to the moderate level (one to two) the rejection rate heats up to the nominal level, which generally indicates the proper size control.

There is an interesting phenomenon for the test power though. In contrast to initial theoretical expectations that too wide bandwidths would average out nonlinear deviations and degrade detection power, experimental results demonstrate preserved high rejection rates even for large bandwidths (up to five). Several reasons might explain this: firstly, the considered nonlinear misspecification, related to the Euclidean norm of the covariates, shows a strong and global smooth deviation from linearity that is not affected by the smoothing effect imposed by a large bandwidth; secondly, the relatively large sample size ($n = 1000$) makes test non-sensitive to the bandwidth, thus preventing power being lost due to kernel oversmoothing; and thirdly, the chosen experimental values for the parameter settings, such as number of bootstrap replications, might be large enough to stabilize the inference.

It is an interesting observation to find the unexpected stability even under big bandwidths and also presents potential for future work. In particular, more theoretical and empirical research is needed to clarify the trade-offs among kernel width, sample size and test power, especially when a signal is weaker or the model misspecification effect is more obvious.

Noted: Some passages of this article have been smoothed and polished with the help of ChatGPT, an A.I. language model created by OpenAI. The final content is the sole responsibility of the authors.

Chapter 7

Python code

```
import numpy as np
from sklearn.metrics.pairwise import rbf_kernel
import matplotlib.pyplot as plt
import seaborn as sns

def generate_data_H0(n, beta, sigma):

    X = np.random.normal(0, 1, size=(n, 5))
    X_matrix = np.hstack([np.ones((n, 1)), X])
    epsilon_real = np.random.normal(0, sigma, size=(n, 1))
    Y = X_matrix @ beta + epsilon_real

    return Y, X, X_matrix, epsilon_real

def generate_data_H1_norm(n, beta, sigma):

    X = np.random.normal(0, 1, size=(n, 5))
    X_matrix = np.hstack([np.ones((n, 1)), X])
    epsilon_real = np.random.normal(0, sigma, size=(n, 1))
    Y = X_matrix @ beta + epsilon_real
    norm_x = np.linalg.norm(X, axis=1, keepdims=True)

    Y_H1 = Y + norm_x
```

```
    return Y_H1, X, X_matrix, epsilon_real
```

```
def generate_data_H1_cos(n, beta, sigma):  
  
    X = np.random.normal(0, 1, size=(n, 5))  
    X_matrix = np.hstack([np.ones((n, 1)), X])  
    epsilon_real = np.random.normal(0, sigma, size=(n, 1))  
    Y = X_matrix @ beta + epsilon_real  
    norm_x = np.linalg.norm(X, axis=1, keepdims=True)  
    non_linear_term = np.cos(norm_x)  
    Y_H1 = Y + non_linear_term  
    return Y_H1, X, X_matrix, epsilon_real
```

```
def generate_data_H1_sin(n, beta, sigma):  
  
    X = np.random.normal(0, 1, size=(n, 5))  
    X_matrix = np.hstack([np.ones((n, 1)), X])  
    epsilon_real = np.random.normal(0, sigma, size=(n, 1))  
    Y = X_matrix @ beta + epsilon_real  
    norm_x = np.linalg.norm(X, axis=1, keepdims=True)  
    non_linear_term = np.sin(norm_x)  
    Y_H1 = Y + non_linear_term  
  
    return Y_H1, X, X_matrix, epsilon_real
```

```
def generate_data_H1_sincos(n, beta, sigma):
```

```

X = np.random.normal(0, 1, size=(n, 5))
X_matrix = np.hstack([np.ones((n, 1)), X])
epsilon_real = np.random.normal(0, sigma, size=(n, 1))
Y = X_matrix @ beta + epsilon_real
norm_x = np.linalg.norm(X, axis=1, keepdims=True)
non_linear_term = np.sin(norm_x) + np.cos(norm_x)
Y_H1 = Y + non_linear_term

return Y_H1, X, X_matrix, epsilon_real

```

```
def generate_data_H1_interaction(n, beta, sigma):
```

```

X = np.random.normal(0, 1, size=(n, 5))
X_matrix = np.hstack([np.ones((n, 1)), X])
epsilon_real = np.random.normal(0, sigma, size=(n, 1))
Y = X_matrix @ beta + epsilon_real
X1_squared = X[:, 0:1]**2
X1_X2 = X[:, 0:1] * X[:, 1:2]
nonlinear_term = X1_squared + X1_X2
Y_H1 = Y + nonlinear_term

return Y_H1, X, X_matrix, epsilon_real

```

```
def kernel(X, sigma_kernel=1):
```

```

gamma = 1 / (2 * sigma_kernel**2)
K = rbf_kernel(X, gamma=gamma)
projection_G = np.linalg.inv(X.T @ X) @ X.T
term2_vec = X @ projection_G @ K

```

```
K_p = K - term2_vec - term2_vec.T + X @ projection_G @ K @ projection_G.T @ X.T

return K, K_p


def ols_via_lstsq(X_matrix, Y):

    beta_hat, _, _, _ = np.linalg.lstsq(X_matrix, Y, rcond=None)
    Y_hat = X_matrix @ beta_hat
    epsilon_hat = Y - Y_hat

    return beta_hat, Y_hat, epsilon_hat


def V_stat(kernel,epsilon):

    n = epsilon.shape[0]
    V = (1 / n) * epsilon.T @ kernel @ epsilon
    return V


def Mult_bootstrap(K_p, epsilon, times, confidence_level):

    n = len(epsilon)
    epsilon = epsilon.flatten()
    V_bootstrap = []
```

```

for _ in range(times):
    W = np.random.normal(0, 1, n)
    weighted_epsilon = W * epsilon
    V_b = (1/n) * weighted_epsilon @ K_p @ weighted_epsilon
    V_bootstrap.append(V_b)

critical_value = np.percentile(V_bootstrap, confidence_level)

return critical_value

def Rejection_rate_H0(repeats, times, confidence_level):

    repeat_vector = np.zeros((repeats, 1))
    V_statistic_vector = np.zeros((repeats, 1))

    for i in range(repeats):
        Y_H0, _, X_matrix_H0, _ = generate_data_H0(n, beta, sigma)
        _, _, epsilon_hat_H0 = ols_via_lstsq(X_matrix_H0, Y_H0)
        _, K_p_H0 = kernel(X_matrix_H0)
        V_Kp_epsilon_hat_H0 = V_stat(K_p_H0, epsilon_hat_H0)
        critical_value = Mult_bootstrap(K_p_H0, epsilon_hat_H0, times, confidence_level)
        repeat_vector[i, 0] = V_Kp_epsilon_hat_H0 > critical_value
        V_statistic_vector[i, 0] = V_Kp_epsilon_hat_H0

    rej_num = np.sum(repeat_vector)
    rej_rate = rej_num / repeats

    return rej_rate, repeat_vector, V_statistic_vector

def Rejection_rate_H1_norm(repeats, times, confidence_level):

```

```

repeat_vector = np.zeros((repeats, 1))

for i in range(repeats):

    Y_H1, _, X_matrix_H1, _ = generate_data_H1_norm(n, beta, sigma)
    _, _, epsilon_hat_H1 = ols_via_lstsq(X_matrix_H1, Y_H1)
    _, K_p_H1 = kernel(X_matrix_H1)
    V_Kp_epsilon_hat_H1 = V_stat(K_p_H1, epsilon_hat_H1)
    critical_value = Mult_bootstrap(K_p_H1, epsilon_hat_H1, times, confidence_level)
    repeat_vector[i, 0] = V_Kp_epsilon_hat_H1 > critical_value

rej_num = np.sum(repeat_vector)
rej_rate = rej_num / repeats
return rej_rate, repeat_vector

def Rejection_rate_H1_sin(repeats, times, confidence_level):

    repeat_vector = np.zeros((repeats, 1))

    for i in range(repeats):
        Y_H1, _, X_matrix_H1, _ = generate_data_H1_norm(n, beta, sigma)
        _, _, epsilon_hat_H1 = ols_via_lstsq(X_matrix_H1, Y_H1)
        _, K_p_H1 = kernel(X_matrix_H1)
        V_Kp_epsilon_hat_H1 = V_stat(K_p_H1, epsilon_hat_H1)
        critical_value = Mult_bootstrap(K_p_H1, epsilon_hat_H1, times, confidence_level)
        repeat_vector[i, 0] = V_Kp_epsilon_hat_H1 > critical_value

    rej_num = np.sum(repeat_vector)
    rej_rate = rej_num / repeats

    return rej_rate, repeat_vector

```

```

def Rejection_rate_H1_cos(repeats, times, confidence_level):

    repeat_vector = np.zeros((repeats, 1))

    for i in range(repeats):
        Y_H1, _, X_matrix_H1, _ = generate_data_H1_norm(n, beta, sigma)
        _, _, epsilon_hat_H1 = ols_via_lstsq(X_matrix_H1, Y_H1)
        _, K_p_H1 = kernel(X_matrix_H1)
        V_Kp_epsilon_hat_H1 = V_stat(K_p_H1, epsilon_hat_H1)
        critical_value = Mult_bootstrap(K_p_H1, epsilon_hat_H1, times, confidence_level)
        repeat_vector[i, 0] = V_Kp_epsilon_hat_H1 > critical_value

    rej_num = np.sum(repeat_vector)
    rej_rate = rej_num / repeats
    return rej_rate, repeat_vector


def Rejection_rate_H1_sincos(repeats, times, confidence_level):

    repeat_vector = np.zeros((repeats, 1))

    for i in range(repeats):
        Y_H1, _, X_matrix_H1, _ = generate_data_H1_sincos(n, beta, sigma)
        _, _, epsilon_hat_H1 = ols_via_lstsq(X_matrix_H1, Y_H1)
        _, K_p_H1 = kernel(X_matrix_H1)
        V_Kp_epsilon_hat_H1 = V_stat(K_p_H1, epsilon_hat_H1)
        critical_value = Mult_bootstrap(K_p_H1, epsilon_hat_H1, times, confidence_level)
        repeat_vector[i, 0] = V_Kp_epsilon_hat_H1 > critical_value

    rej_num = np.sum(repeat_vector)
    rej_rate = rej_num / repeats
    return rej_rate, repeat_vector

```

```
def Rejection_rate_H1_interaction(repeats, times, confidence_level):

    repeat_vector = np.zeros((repeats, 1))
    for i in range(repeats):
        Y_H1, _, X_matrix_H1, _ = generate_data_H1_interaction(n, beta, sigma)
        _, _, epsilon_hat_H1 = ols_via_lstsq(X_matrix_H1, Y_H1)
        _, K_p_H1 = kernel(X_matrix_H1)
        V_Kp_epsilon_hat_H1 = V_stat(K_p_H1, epsilon_hat_H1)
        critical_value = Mult_bootstrap(K_p_H1, epsilon_hat_H1, times, confidence_level)
        repeat_vector[i, 0] = V_Kp_epsilon_hat_H1 > critical_value

    rej_num = np.sum(repeat_vector)
    rej_rate = rej_num / repeats
    return rej_rate, repeat_vector
```

Bibliography

- [1] H. J. BIERENS, *Consistent model specification tests*, Journal of Econometrics, 20 (1982), pp. 105–134.
- [2] S. BOCHNER, *Über die fourier-stieltjes transformationen*, Mathematische Zeitschrift, 36 (1932), pp. 1–12.
- [3] M. A. DELGADO, M. A. DOMÍNGUEZ, AND P. LAVERGNE, *Consistent tests of conditional moment restrictions*, Annales d’Économie et de Statistique, (2006), pp. 33–67.
- [4] L. P. HANSEN, *Large sample properties of generalized method of moments estimators*, Econometrica: Journal of the econometric society, (1982), pp. 1029–1054.
- [5] M. A. HEARST, S. T. DUMAIS, E. OSUNA, J. PLATT, AND B. SCHOLKOPF, *Support vector machines*, IEEE Intelligent Systems and their applications, 13 (1998), pp. 18–28.
- [6] Y. LI AND X. SONG, *Consistent test for conditional moment restriction models in reproducing kernel hilbert spaces*, (2022).
- [7] K. MUANDET, W. JITKRITTUM, AND J. KÜBLER, *Kernel conditional moment test via maximum moment restriction*, in Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI), J. Peters and D. Sontag, eds., vol. 124 of Proceedings of Machine Learning Research, PMLR, 03–06 Aug 2020, pp. 41–50.
- [8] W. K. NEWEY, *Maximum likelihood specification testing and conditional moment tests*, Econometrica: Journal of the Econometric Society, (1985), pp. 1047–1070.
- [9] R. J. SERFLING, *Approximation theorems of mathematical statistics*, John Wiley & Sons, 2009.
- [10] M. B. STINCHCOMBE AND H. WHITE, *Consistent specification testing with nuisance parameters present only under the alternative*, Econometric theory, 14 (1998), pp. 295–325.
- [11] G. TAUCHEN, *Diagnostic testing and evaluation of maximum likelihood models*, Journal of Econometrics, 30 (1985), pp. 415–443.